

Computer Vision Study Guide

CO2 concepts — image enhancement & segmentation. Use this to understand each technique, then write the test answers yourself, in your own words.

1. Frequency Domain Basics

An image can be represented in the frequency domain via the Fourier Transform. Low frequencies correspond to slowly-varying intensity — flat backgrounds and smooth shading. High frequencies correspond to rapid intensity changes — edges, fine texture, and noise. Frequency-domain filtering multiplies the transformed image by a filter function $H(u,v)$ and then transforms it back to the spatial domain.

2. Low-Pass Filtering — smoothing and noise removal

A low-pass filter keeps low-frequency components and attenuates high-frequency ones. Since random noise is concentrated in the high frequencies, suppressing them reduces noise while leaving the image's overall smooth structure intact — this is also how you get a deliberately smoothed/blurred version of an image.

- **Ideal low-pass filter (ILPF):** a sharp cutoff at radius D_0 — everything inside passes, everything outside is blocked. Simple, but the sharp cutoff causes ringing artifacts in the spatial domain.
- **Butterworth low-pass filter:** a smoother transition controlled by an order n . Reduces ringing compared to the ideal filter, giving a middle ground between smoothing strength and artifact control.
- **Gaussian low-pass filter:** a smooth, bell-shaped roll-off with no ringing at all. It's the most commonly used in practice because the transition is gradual.

Effect on image quality: as the cutoff frequency is lowered, noise and fine detail both drop out together — the image gets smoother but also loses sharpness, so there's a direct trade-off between noise suppression and detail preservation.

3. High-Pass Filtering — edge enhancement

A high-pass filter does the opposite: it attenuates low frequencies (the smooth background) and passes high frequencies (edges and fine detail). Since edges are rapid intensity transitions, boosting the high frequencies makes them stand out more sharply. Ideal, Butterworth, and Gaussian high-pass versions exist too — each is essentially 1 minus the matching low-pass filter, so the same ringing-vs-smoothness trade-offs apply.

Effect on image quality: edges and fine structures become more pronounced and the image looks sharper, but flat regions can look darker/flatter, and because noise is also high-frequency, it gets amplified along with the edges.

4. Intensity-Based Segmentation — thresholding

When an object's intensity is clearly different from the background, global thresholding is the natural approach: pick a threshold T , and classify every pixel above T as foreground and below T as background (or vice versa).

- **Otsu's method:** automatically chooses T from the image histogram by finding the value that best separates two classes — minimizing the spread within each class while maximizing the separation between them. It works best when the histogram is bimodal (a clear peak for the object and another for the background).

Effectiveness: excellent under uniform lighting and clear contrast; it struggles when illumination is uneven or when foreground and background intensities overlap.

5. Region-Based Segmentation

Region growing starts from one or more seed pixels and repeatedly adds neighbouring pixels that satisfy a similarity criterion — typically, an intensity difference under some threshold. The region keeps expanding until no remaining neighbour qualifies. This makes it a natural fit whenever the goal is to group adjacent pixels that already share similar intensity, since that similarity is exactly the rule driving growth.

6. Edge Detection

Edge detection looks for points where intensity changes sharply. Gradient operators such as Sobel or Prewitt convolve the image with small masks approximating the horizontal and vertical derivatives; the gradient magnitude highlights the edges.

The Canny detector is a fuller pipeline: Gaussian smoothing to cut noise, gradient computation, non-maximum suppression to thin edges to single-pixel width, then double thresholding with hysteresis — strong edges are kept, and weak responses are kept only if they connect to a strong edge, so isolated noise gets dropped. This generally gives cleaner, thinner, better-connected edges than a plain gradient operator, at the cost of more computation.

7. Fixing Broken Edges — edge linking

Gaps in a detected edge map usually come from noise or a locally weak gradient along a real edge. Common fixes:

- Morphological dilation/closing — thickens edge fragments so small gaps get bridged.
- Hough transform — detects lines (or other shapes) by voting in a parameter space, effectively filling in a straight edge even where individual pixels are missing.
- Hysteresis-based tracking (as used in Canny) — explicitly links weak-edge pixels to already-confirmed strong edges when they are neighbours.

These steps turn a fragmented, noisy edge map into continuous boundaries that later steps (like contour extraction) can actually use.

8. Extracting Object Outlines — boundary/contour extraction

Once a region is segmented (e.g. via thresholding) or an edge map is available, boundary-following algorithms trace the outline of each region pixel by pixel — Moore-neighbour tracing is a common example — recording the sequence of boundary points, sometimes compactly as a chain code. The result is a clean outline of the object, useful for shape description or further analysis.

Quick reference

- Frequency-domain low-pass filtering → noise removal, smoothing
- Frequency-domain high-pass filtering → edge enhancement
- Global thresholding / Otsu → separating foreground and background by intensity
- Region growing → grouping similar adjacent pixels
- Sobel / Canny → detecting edges
- Morphological linking / Hough transform → repairing broken edges
- Boundary tracing / contour extraction → extracting object outlines